

EXPERIMENT – 3

IMPLEMENTING ASSET MANAGEMENT WITH HYPERLEDGER FABRIC

Aim

To implement an Asset Management application using Hyperledger Fabric by creating, reading, updating, transferring, and deleting assets through JavaScript chaincode and verifying the transactions on the blockchain ledger.

Software Requirements

- Kali Linux / Ubuntu
- Docker
- Docker Compose
- Node.js (Version 18 or later)
- npm
- Git
- Hyperledger Fabric Samples
- Hyperledger Fabric Binaries
- Hyperledger Fabric Docker Images
- Visual Studio Code (Optional)

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: Minimum 8 GB
- Storage: Minimum 20 GB Free Disk Space
- Internet Connection

Step 1: Navigate to the Fabric Test Network

```
cd ~/fabric-dev/fabric-samples/test-network
```

```
dharsna@DHARSHU:~$ cd ~/fabric-dev/fabric-samples/test-network
dharsna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ ./network.sh up createChannel -ca
```

Step 2: Start the Hyperledger Fabric Network

```
./network.sh up createChannel -ca
```

```
2026-09-08 15:16:10.717 UTC 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
2026-09-08 15:16:10.736 UTC 0002 INFO [channelCmd] update -> Successfully submitted channel update
Anchor peer set for org 'Org2MSP' on channel 'mychannel'
Channel 'mychannel' joined
dharsna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ |
```

Step 3: Deploy the Asset Transfer Chaincode

```
./network.sh deployCC \
-ccl javascript \
-ccn basic \
-ccp ../asset-transfer-basic/chaincode-javascript
```

```
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]
Query chaincode definition successful on peer0.org1 on channel 'mychannel'
Using organization 2
Querying chaincode definition on peer0.org2 on channel 'mychannel'...
Attempting to Query committed status on peer0.org2, Retry after 3 seconds.
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic
+ res=0
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]
Query chaincode definition successful on peer0.org2 on channel 'mychannel'
Chaincode initialization is not required
dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$
```

Step 4: Initialize the Ledger

```
peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile $ORDERER_CA \
-C mychannel \
-n basic \
--peerAddresses localhost:7051 \
--tlsRootCertFiles $PEER0_ORG1_CA \
--peerAddresses localhost:9051 \
--tlsRootCertFiles $PEER0_ORG2_CA \
-c '{"function":"InitLedger","Args":[]}'
```

```
dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile $ORDERER_CA \
-C mychannel \
-n basic \
--peerAddresses localhost:7051 \
--tlsRootCertFiles $PEER0_ORG1_CA \
--peerAddresses localhost:9051 \
--tlsRootCertFiles $PEER0_ORG2_CA \
-c '{"function":"InitLedger","Args":[]}'
2026-09-08 15:19:00.627 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$
```

Step 5: Query All Assets

```
peer chaincode query \
-C mychannel \
-n basic \
-c '{"Args":["GetAllAssets"]}'
```

```
dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ peer chaincode query \
-C mychannel \
-n basic \
-c '{"Args":["GetAllAssets"]}'
[{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5,"docType":"asset"}, {"AppraisedValue":400,"Color":"red","ID":"asset2","Owner":"Brad","Size":5,"docType":"asset"}, {"AppraisedValue":500,"Color":"green","ID":"asset3","Owner":"Jin Soo","Size":10,"docType":"asset"}, {"AppraisedValue":600,"Color":"yellow","ID":"asset4","Owner":"Max","Size":10,"docType":"asset"}, {"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15,"docType":"asset"}, {"AppraisedValue":800,"Color":"white","ID":"asset6","Owner":"Michel","Size":15,"docType":"asset"}]
dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$
```

Step 6: Create a New Asset

```
peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile $ORDERER_CA \
-C mychannel \
-n basic \
--peerAddresses localhost:7051 \
--tlsRootCertFiles $PEER0_ORG1_CA \
--peerAddresses localhost:9051 \
--tlsRootCertFiles $PEER0_ORG2_CA \
-c '{"function":"CreateAsset","Args":["asset11","White","12","Arun","900"]}'
```

```
dharshta@DHARSHU:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile $ORDERER_CA \
-C mychannel \
-n basic \
--peerAddresses localhost:7051 \
--tlsRootCertFiles $PEER0_ORG1_CA \
--peerAddresses localhost:9051 \
--tlsRootCertFiles $PEER0_ORG2_CA \
-c '{"function":"CreateAsset","Args":["asset11","White","12","Arun","900"]}'
2026-09-08 15:19:56.070 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200 payload:{"ID":"asset11","Color":"White","Size":12,"Owner":"Arun","AppraisedValue":900}
dharshta@DHARSHU:~/fabric-dev/fabric-samples/test-network$
```

Step 7: Read the Asset Details

```
peer chaincode query \
-C mychannel \
-n basic \
-c '{"Args":["ReadAsset","asset11"]}'
```

```
dharshta@DHARSHU:~/fabric-dev/fabric-samples/test-network$ peer chaincode query \
-C mychannel \
-n basic \
-c '{"Args":["ReadAsset","asset11"]}'
{"AppraisedValue":900,"Color":"White","ID":"asset11","Owner":"Arun","Size":12}
dharshta@DHARSHU:~/fabric-dev/fabric-samples/test-network$
```

Step 8: Update Asset Information

```
peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile $ORDERER_CA \
-C mychannel \
-n basic \
--peerAddresses localhost:7051 \
--tlsRootCertFiles $PEER0_ORG1_CA \
--peerAddresses localhost:9051 \
--tlsRootCertFiles $PEER0_ORG2_CA \
-c '{"function":"UpdateAsset","Args":["asset11","Black","15","Arun","1200"]}'
```

```

dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile $ORDERER_CA \
-C mychannel \
-n basic \
--peerAddresses localhost:7051 \
--tlsRootCertFiles $PEER0_ORG1_CA \
--peerAddresses localhost:9051 \
--tlsRootCertFiles $PEER0_ORG2_CA \
-c '{"function":"UpdateAsset","Args":["asset11","Black","15","Arun","1200"]}'
2026-09-08 15:21:34.355 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200 payload:{"type":"Buffer","data":[]}
dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$

```

Step 9: Transfer Asset Ownership

```

peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile $ORDERER_CA \
-C mychannel \
-n basic \
--peerAddresses localhost:7051 \
--tlsRootCertFiles $PEER0_ORG1_CA \
--peerAddresses localhost:9051 \
--tlsRootCertFiles $PEER0_ORG2_CA \
-c '{"function":"TransferAsset","Args":["asset11","David"]}'

```

```

dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile $ORDERER_CA \
-C mychannel \
-n basic \
--peerAddresses localhost:7051 \
--tlsRootCertFiles $PEER0_ORG1_CA \
--peerAddresses localhost:9051 \
--tlsRootCertFiles $PEER0_ORG2_CA \
-c '{"function":"TransferAsset","Args":["asset11","David"]}'
2026-09-08 15:21:57.370 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200 payload:{"type":"Buffer","data":[]}
dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$

```

Step 10: Verify the Updated Asset

```

peer chaincode query \
-C mychannel \
-n basic \
-c '{"Args":["ReadAsset","asset11"]}'

```

```

dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ peer chaincode query \
-C mychannel \
-n basic \
-c '{"Args":["ReadAsset","asset11"]}'
{"AppraisedValue":1200,"Color":"Black","ID":"asset11","Owner":"David","Size":"15"}
dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$

```

Step 11: Delete the Asset

```

peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile $ORDERER_CA \

```

```
-C mychannel \
-n basic \
--peerAddresses localhost:7051 \
--tlsRootCertFiles $PEER0_ORG1_CA \
--peerAddresses localhost:9051 \
--tlsRootCertFiles $PEER0_ORG2_CA \
-c '{"function":"DeleteAsset","Args":["asset11"]}'
```

```
dharsbna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile $ORDERER_CA \
-C mychannel \
-n basic \
--peerAddresses localhost:7051 \
--tlsRootCertFiles $PEER0_ORG1_CA \
--peerAddresses localhost:9051 \
--tlsRootCertFiles $PEER0_ORG2_CA \
-c '{"function":"DeleteAsset","Args":["asset11"]}'
2026-09-08 15:22:53.486 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200 payload:{"type":"","Buffer":"","data":[]}
```

Step 12: Verify Asset Deletion

```
peer chaincode query \
-C mychannel \
-n basic \
-c '{"Args":["ReadAsset","asset11"]}'
```

```
dharsbna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ peer chaincode query \
-C mychannel \
-n basic \
-c '{"Args":["ReadAsset","asset11"]}'
Error: endorsement failure during query. response: status:500 message:"The asset asset11 does not exist"
```

Step 13: Stop the Network

```
./network.sh down
```

```
dharsbna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ ./network.sh down
Using docker and docker-compose
Stopping network
[+] down 10/10
✓Container ca_org2 Removed 10.7s
✓Container orderer.example.com Removed 0.5s
✓Container ca_org1 Removed 10.4s
✓Container peer0.org1.example.com Removed 1.0s
✓Container peer0.org2.example.com Removed 1.2s
✓Container ca_orderer Removed 10.6s
✓Volume compose_peer0.org2.example.com Removed 0.0s
✓Volume compose_peer0.org1.example.com Removed 0.0s
✓Volume compose_orderer.example.com Removed 0.0s
✓Network fabric_test Removed 0.2s
Error response from daemon: get docker_orderer.example.com: no such volume
Error response from daemon: get docker_peer0.org1.example.com: no such volume
Error response from daemon: get docker_peer0.org2.example.com: no such volume
Removing remaining containers
Removing generated chaincode docker images
Untagged: dev-peer0.org2.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5f5b664c87b24c035bee15cbcc7f59b629-c68f8ecc2a0ff0cc14976
15f5e33a8bfa908e0fd3555f864cfa1398b52d341cb:latest
Deleted: sha256:ab81c0a79afe311012713e2e6bbb01f7a6e0cc82cd96b13feb0eedb44f77fbca
Untagged: dev-peer0.org1.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5f5b664c87b24c035bee15cbcc7f59b629-d3cf9b1f4b9747cc8433a
c0e74bfaade1cf0b97f5c1163835f604aa885d60dc:latest
Deleted: sha256:ca2660e86a9bfc8513c793d490cc42f57a1cc9e8c8a343cd3f56518afe79faff
```

Result

The Asset Management application was successfully implemented using Hyperledger Fabric. The asset was created, read, updated, transferred to another owner, and deleted using JavaScript chaincode, and the transactions were verified on the blockchain ledger.